Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

Oliver James and **Sungho Hong**†
Center for Cognition and Sociality, Institute for Basic Science; † sunghohong@ibs.re.kr

Introduction

1. Modeling **large-scale, biophysically detailed neural networks** increasingly relies on **high-performance computing (HPC)**.

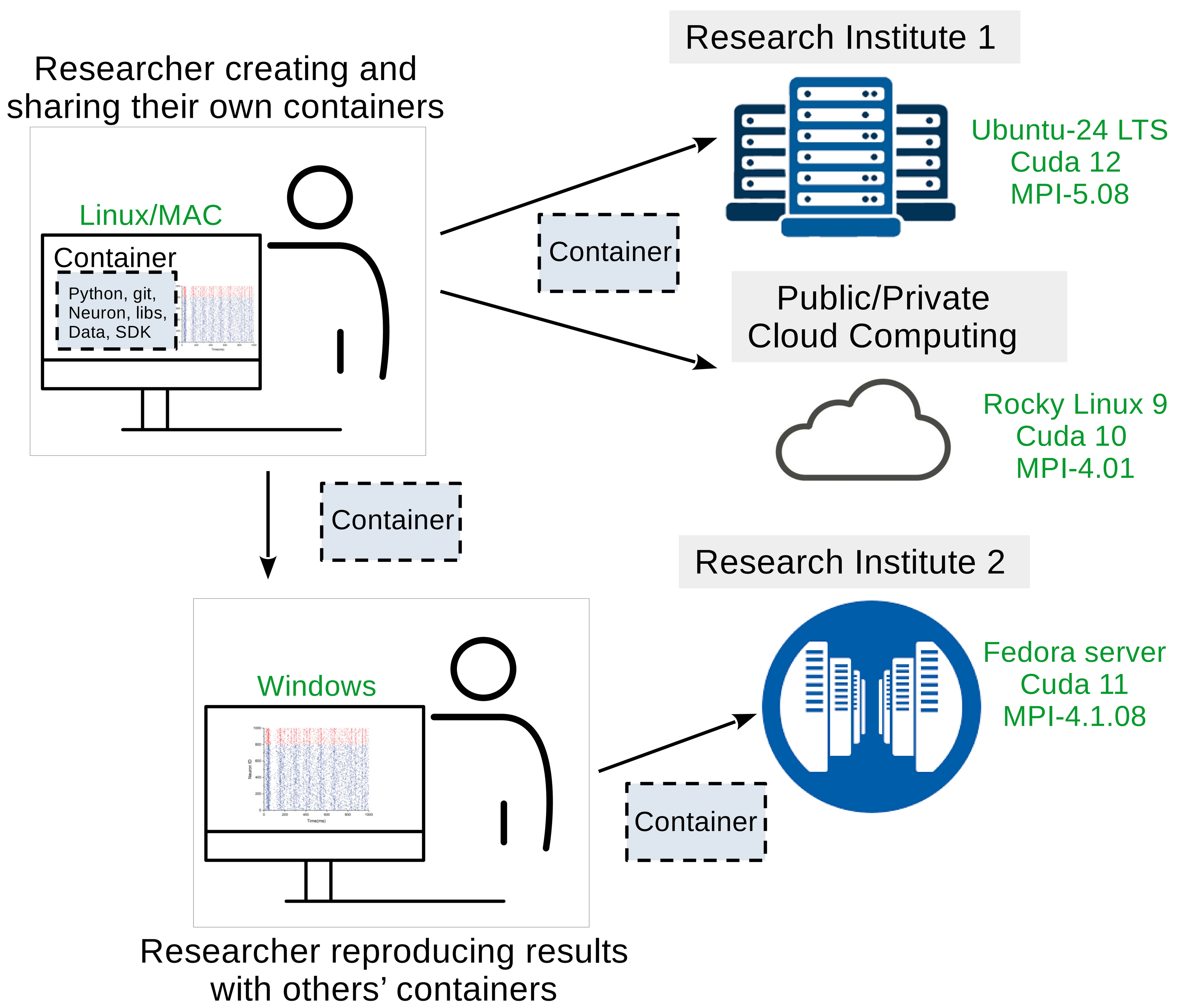
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to **software incompatibilities** and **complex dependencies**.

3. We address this issue using **Singularity containers** that encapsulats the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.

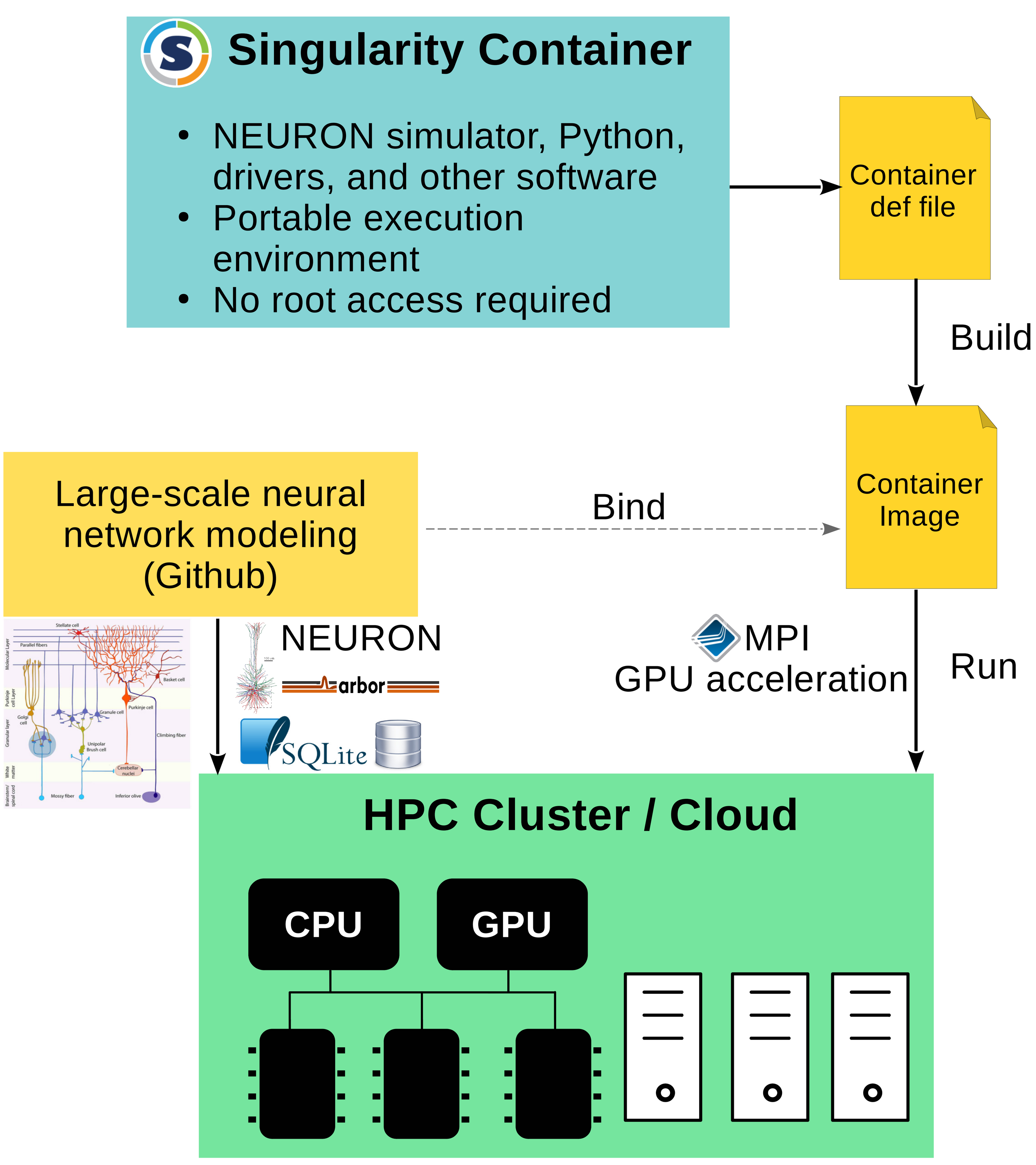
4. We give practical demonstrations by benchmarking HPC environments with **cerebellar mossy fiber—granule cell network** simulations.

Containerization for Scientific Computing

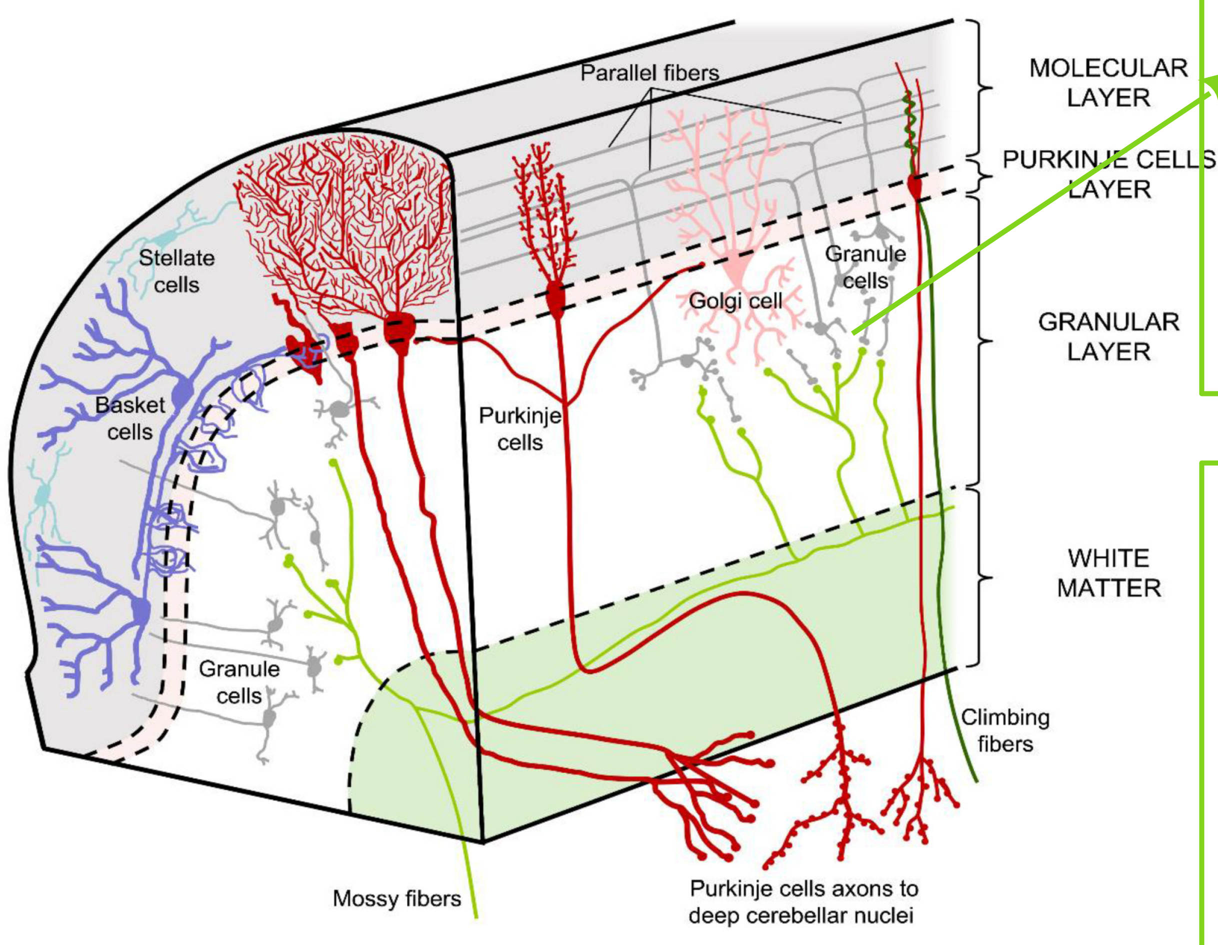
A container is a lightweight, standalone, virtualized executable environment that includes all dependencies required to run large-scale scientific models.



Our Computational Framework



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



EM Data

MF - GRC Connectivity

Nguyen, T.M., Thomas, L.A., Rhoades, J.L. et al. Structured cerebellar connectivity supports resilient pattern separation. **Nature** 613, 543–549 (2023)

MF → 1042 cells
GRC → 3953 cells

Cell Models

MF → spike generators

GRC → Bio-physically detailed model with 12 ion-channel mechanisms

Sudhakar, S.K., Hong, S., et.al., 2017. Spatiotemporal network coding of physiological mossy fiber inputs by cerebellar granular layer. **PLOS Compu. Biol** 13

We run multiple copies ($n = 280$) of the network connectivity model simultaneously to simulate extreme computational workloads.

This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

Table: Network Configuration

Network elements	Count (millions)
MF	0.299
GRC	1.099
Compartments	3.297
Synapses	3.468

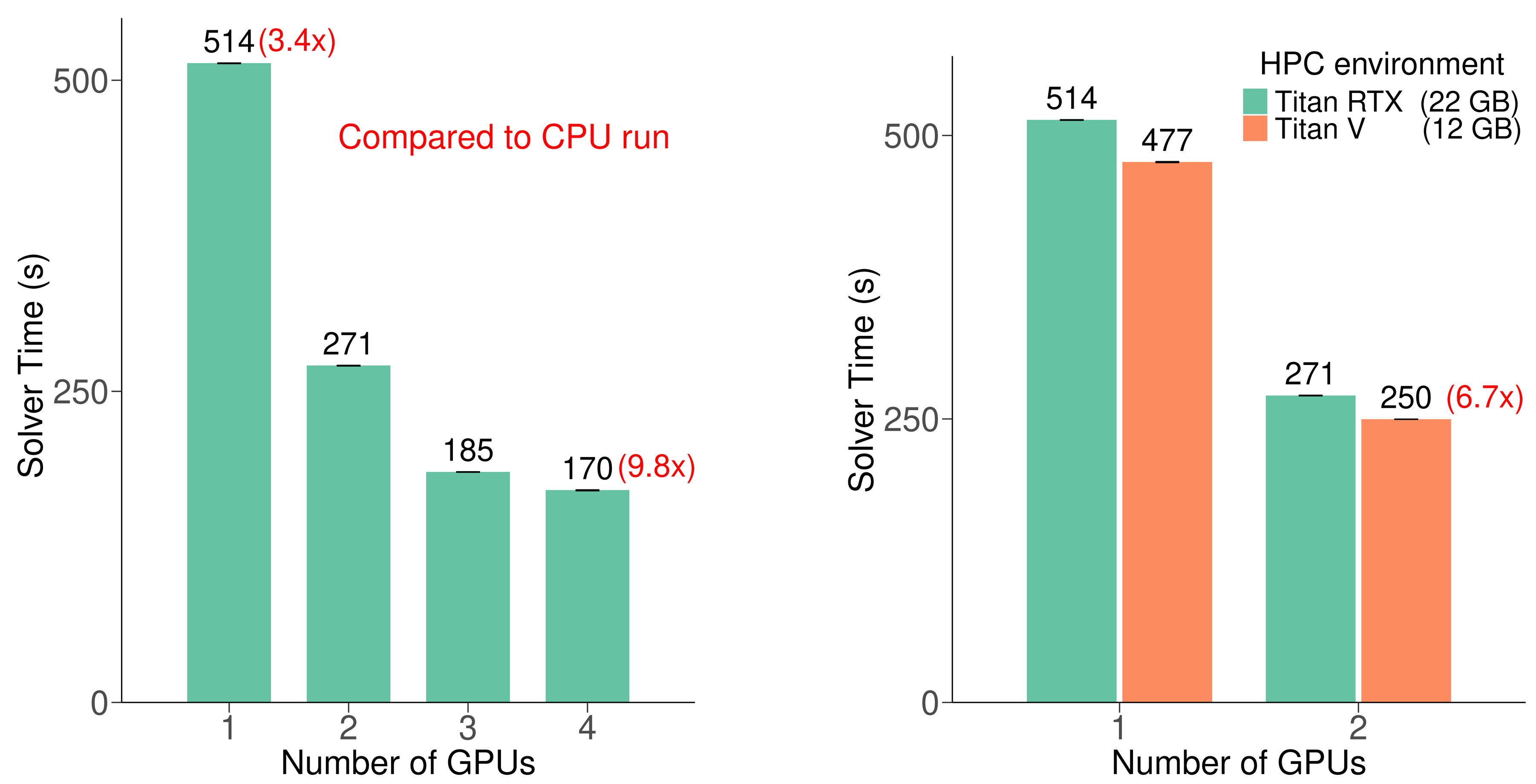
Table: Simulation Setup

Simulation time	1 s
Simulation software	CoreNEURON

Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	1670	CoreNEURON-CPU



(a) TITAN RTX architecture: Solver time vs. varying numbers of GPUs in CORENeuron-GPU simulator.

(b) Comparison of solver time between HPC environments

TITAN RTXs outperform 30-core CPUs by up to 9.8x, with performance improving as GPU count increases.

Future works:

☞ **Develop a biologically realistic cerebellar circuit model** including granule layer (with Golgi cells), Purkinje layer, and other interneurons.

☞ **Benchmark network performance across multiple institutions** to evaluate and understand the scalability and performance limits of HPC systems.

Acknowledgment

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